Topic 12: Acid-base Equilibria

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include a series of pH titrations to determine titration curves, calculating an ionisation constant for a pH indicator, making up and investigating the properties of buffer solutions and using a pH meter.

Mathematical skills that could be developed in this topic include the use of logarithms and exponentials for converting from concentration to pH and *vice versa*, rearranging K_a expressions into expressions suitable for calculating pH of a buffer solution, plotting and interpreting titration curves.

Within this topic, students can consider how the historical development of theories explaining acid and base behaviour show that scientific ideas change as a result of new evidence and fresh thinking. They can also relate their study of buffer solutions to a range of applications in living cells, medicines, foods and the natural environment.

Students should:
1. know that a Brønsted–Lowry acid is a proton donor and a Brønsted–Lowry base is a proton acceptor
2. know that acid-base reactions involve the transfer of protons
3. be able to identify Brønsted–Lowry conjugate acid-base pairs
4. be able to define the term 'pH'
5. be able to calculate pH from hydrogen ion concentration
6. be able to calculate the concentration of hydrogen ions, in mol dm^{-3}, in a solution from its pH, using the expression $[H^+] = 10^{-pH}$
7. understand the difference between a strong acid and a weak acid in terms of degree of dissociation
8. be able to calculate the pH of a strong acid
9. be able to deduce the expression for the acid dissociation constant, K_a, for a weak acid and carry out relevant calculations
10. be able to calculate the pH of a weak acid making relevant assumptions
11. be able to define the ionic product of water, K_w
12. be able to calculate the pH of a strong base from its concentration, using K_w
13. be able to define the terms 'pK_a' and 'pK_w'
14. be able to analyse data from the following experiments: i measuring the pH of a variety of substances, e.g. equimolar solutions of strong and weak acids, strong and weak bases, and salts ii comparing the pH of a strong acid and a weak acid after dilution 10, 100 and 1000 times

Students should:
16. be able to draw and interpret titration curves using all combinations of strong and weak monobasic acids and bases
17. be able to select a suitable indicator, using a titration curve and appropriate data
18. know what is meant by the term 'buffer solution'
19. understand the action of a buffer solution
20. be able to calculate the pH of a buffer solution given appropriate data
21. be able to calculate the concentrations of solutions required to prepare a buffer solution of a given pH
22. understand how to use a weak acid–strong base titration curve to: i demonstrate buffer action ii determine K_a from the pH at the point where half the acid is neutralised
23. understand why there is a difference in enthalpy changes of neutralisation values for strong and weak acids
24. understand the roles of carbonic acid molecules and hydrogencarbonate ions in controlling the pH of blood
CORE PRACTICAL 9: Finding the K_a value for a weak acid